Exact Typed Phase Chains with Recomputable Route Closure

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Abstract

This paper defines a fixed typed phase chain over the canonical sparse residual tensor boundary. The route takes a dense input and a second canonical operation tensor, crosses an exact opaque transport boundary, performs typed calculation, crosses back, restores the selected result, and atomically manifests those exact bytes. Every transition fixes its epoch, position, input and output type, operator, evaluator, predecessor, and canonical value commitments. Exact continuity is established from canonical typed bytes; hashes provide separate mutation-detection evidence. Verification is streaming and retains no vector of phase values or alternate mapback. A bounded active-window construction remains an independent tensor-realization section rather than an invented bridge into dense restoration. The result is one exact route result or typed rejection.

Keywords: typed phase chain; route closure; canonical tensor; exact composition; streaming verification; fail-closed evidence.

Contents

1 Introduction **3**

2 Exact imported signatures **3**

3 Typed main route **4**

4 Independent bounded tensor section **6**

5 Optional proof section **6**

6 Canonical identities and exact equality **6**

7 Link construction **7**

8 Streaming verification **7**

9 Presently verified section re-entry **8**

10 Datatype and aperture boundary **8**

11 Patent foundation locators **8**

12 Main theorem **9**

13 Failure matrix **9**

14 Position relative to established constructions **9**

15 Scope **10**

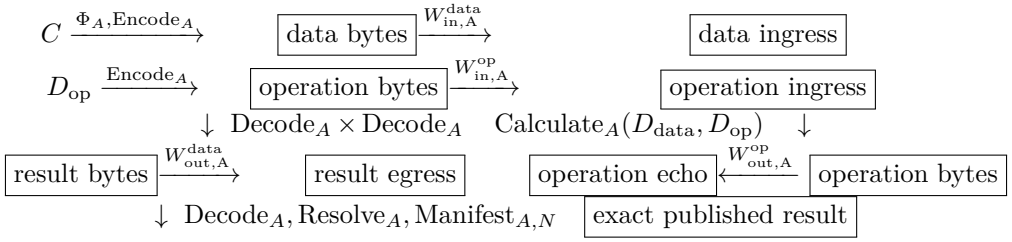
16 Conclusion **10**

A Candidate formal specification **10**

1 Introduction

An ordered list of stage names is not a typed route. Exact composition requires each stage to consume precisely the value produced by its predecessor, including rejection type and epoch. It also requires the canonical calculation operand to remain data rather than becoming an outer opcode, and it requires present verification instead of a stored success flag.

The construction below fixes one main route and one independent bounded tensor-realization section. The main lifecycle is:



Both canonical operands cross complete, separately declared Wire routes. The operation route returns an exact unchanged echo; no calculation operand travels beside or around an opaque crossing.

Origamold supplies a bounded realization of a selected tensor closure. Its codomain is a tensor window, not dense restored bytes; it is therefore bound as an independent section rather than inserted between Resolve and manifestation. This distinction is what makes the cross-paper composition type-correct.

The patent supplies the envelope foundation and fidelity/aperture clamps [9]. Ionizer supplies the canonical tensor, exact inverse, selector, and two-operand calculation boundary [11]. Papers II–IV supply exact transport, bounded tensor realization, and atomic manifestation respectively [10, 8, 7]. The internal method of a calculation engine is outside this paper.

2 Exact imported signatures

Fix a valid Ionizer epoch A. Let

$$\mathcal{C}_A = U^*, \qquad \mathcal{D}_A = \mathrm{CanonicalTensor}_A, \qquad \mathcal{B}_A = \mathrm{Image}(\mathrm{Encode}_A).$$

Set $\mathrm{Encode}_A = \mathrm{CanonicalEncoding}_A$ and $\mathrm{Decode}_A = \mathrm{CanonicalDecode}_A$. The imported maps are

$$\Phi_A : \mathcal{C}_A \to \mathcal{D}_A, \tag{1}$$
$$\mathrm{Encode}_A : \mathcal{D}_A \to \mathcal{B}_A, \tag{2}$$
$$\mathrm{Decode}_A : \mathbb{B}^* \to \mathcal{D}_A \sqcup \{\mathsf{REJECT}\}, \tag{3}$$
$$\mathrm{Calculate}_A : \mathcal{D}_A \times \mathcal{D}_A \to \mathcal{D}_A \sqcup \{\mathsf{REJECT}\}, \tag{4}$$
$$\mathrm{Resolve}_A : \mathcal{D}_A \times \mathrm{Selector}_A \to \mathcal{C}_A \sqcup \{\mathsf{REJECT}\}. \tag{5}$$

In particular, Resolve returns bytes. The history-valued Ionizer operator is μ_A; this paper does not rename its codomain.

Paper II supplies the exact owner signatures

$$W_{\mathrm{in},A}(b) = (b_{\mathrm{in}}, \rho, d_{\mathrm{in}}), \tag{6}$$
$$W_{\mathrm{out},A}(\rho, d_{\mathrm{in}}, b_{\mathrm{out}}) = (b_{\mathrm{return}}, d_{\mathrm{out}}). \tag{7}$$

The theorem is instantiated twice as two complete routes with data projections

$$W^{\text{data}}_{\text{in},A}(b_d) = W_{\text{in},A}(b_d) = (b_d, \rho_d, d_{d,\text{in}}), \tag{8}$$

$$W^{\text{data}}_{\text{out},A}(\rho_d, d_{d,\text{in}}, b_r) = W_{\text{out},A}(\rho_d, d_{d,\text{in}}, b_r) = (b_r, d_{d,\text{out}}), \tag{9}$$

$$W^{\text{op}}_{\text{in},A}(b_o) = W_{\text{in},A}(b_o) = (b_o, \rho_o, d_{o,\text{in}}), \tag{10}$$

$$W^{\text{op}}_{\text{out},A}(\rho_o, d_{o,\text{in}}, b_o) = W_{\text{out},A}(\rho_o, d_{o,\text{in}}, b_o) = (b_{o,\text{echo}}, d_{o,\text{out}}), \tag{11}$$

or typed rejection, with $b_{o,\text{echo}} = b_o$ byte for byte on success. The data and operation route states are separate surrounding products, not additional fields in either owner tuple. The two route/digest states are distinct and both lifecycles close. The Origamold section uses demand $Q = (q, b)$ and profile R_A:

$$\text{Realize}_A(D, Q; R_A) = (W_q, K_q), \qquad \text{Read}(W_q) = \beta_{A,\Lambda}(D|q),$$

or typed rejection, where $\text{ConformingProfile}_A(R_A)$ and $0 \le |\beta_{A,\Lambda}(D|q)| \le b \le B_{\text{window}}$. The Energizer boundary is

$$\text{Manifest}_{A,N}(D, s, t) = \text{PUBLISHED}(t, |C_s|, H(C_s), B^*, \text{epoch}(D), s)$$

or rejection, where $C_s = \text{Resolve}_A(D, s)$. No companion signature uses a "selected history to window to manifestation" chain.

3 Typed main route

Definition 1 (Route)**.** The main route is the fixed vector

$$\begin{aligned} R_A = (&\mathsf{IONIZE}, \mathsf{ENCODE_DATA}, \mathsf{WIRE_DATA_INGRESS}, \\ &\mathsf{ENCODE_OPERATION}, \mathsf{WIRE_OPERATION_INGRESS}, \mathsf{DECODE_OPERANDS}, \\ &\mathsf{CALCULATE}, \mathsf{ENCODE_RESULT}, \mathsf{WIRE_DATA_EGRESS}, \\ &\mathsf{WIRE_OPERATION_EGRESS}, \mathsf{DECODE_RESULT}, \mathsf{RESOLVE}, \mathsf{MANIFEST}, \mathsf{CLOSE}). \end{aligned}$$

This vector is the only admitted main route. A vector containing FAKE, BOGUS, a renamed position, an insertion, deletion, duplication, or reordering is rejected even when its adjacent types happen to match.

For route input (C, D_{op}, s, t), the adjacent types are listed in Table 1. Wire evidence consists only of fixed route and frame digests; it is not semantic input to calculation.

Table 1: Exact phase types.

Boundary	Canonical product type
X_0	$\mathcal{C}_A \times \mathcal{D}_A \times S_A \times T_N$
X_1	$\mathcal{D}_A \times \mathcal{D}_A \times S_A \times T_N$
X_2	$\mathcal{B}_A \times \mathcal{D}_A \times S_A \times T_N$
X_3	$\mathcal{B}_A \times \mathcal{D}_A \times S_A \times T_N \times E^d_{\text{in}}$
X_4	$\mathcal{B}_A \times \mathcal{B}_A \times S_A \times T_N \times E^d_{\text{in}}$
X_5	$\mathcal{B}_A^2 \times S_A \times T_N \times E^d_{\text{in}} \times E^o_{\text{in}}$
X_6	$\mathcal{D}_A^2 \times \mathcal{B}_A \times S_A \times T_N \times E^d_{\text{in}} \times E^o_{\text{in}}$
X_7	$\mathcal{D}_A \times \mathcal{B}_A \times S_A \times T_N \times E^d_{\text{in}} \times E^o_{\text{in}}$
X_8	$\mathcal{B}_A^2 \times S_A \times T_N \times E^d_{\text{in}} \times E^o_{\text{in}}$
X_9	$\mathcal{B}_A^2 \times S_A \times T_N \times E^d_{\text{out}} \times E^o_{\text{in}}$
X_{10}	$\mathcal{B}_A^2 \times S_A \times T_N \times E^d_{\text{out}} \times E^o_{\text{out}}$
X_{11}	$\mathcal{D}_A \times S_A \times T_N$
X_{12}	$\mathcal{D}_A \times S_A \times T_N \times \mathcal{C}_A$
X_{13}	$\text{ManifestReceipt}_{A,N}$
X_{14}	$\text{RouteResult}_{A,N}$

The phase equations are:

$$x_1 = (\Phi_A(C), D_{\text{op}}, s, t), \tag{12}$$
$$x_2 = (\text{Encode}_A(D_{\text{data}}), D_{\text{op}}, s, t), \tag{13}$$
$$x_3 = (b_d, D_{\text{op}}, s, t, (\rho_d, d_{d,\text{in}})), \tag{14}$$
$$x_4 = (b_d, \text{Encode}_A(D_{\text{op}}), s, t, (\rho_d, d_{d,\text{in}})), \tag{15}$$
$$x_5 = (b_d, b_o, s, t, (\rho_d, d_{d,\text{in}}), (\rho_o, d_{o,\text{in}})), \tag{16}$$
$$x_6 = (\text{Decode}_A(b_d), \text{Decode}_A(b_o), b_o, s, t, E^d_{\text{in}}, E^o_{\text{in}}), \tag{17}$$
$$x_7 = (\text{Calculate}_A(D_{\text{data}}, D_{\text{op}}), b_o, s, t, E^d_{\text{in}}, E^o_{\text{in}}), \tag{18}$$
$$x_8 = (\text{Encode}_A(D_{\text{result}}), b_o, s, t, E^d_{\text{in}}, E^o_{\text{in}}), \tag{19}$$
$$x_9 = (b_r, b_o, s, t, E^d_{\text{out}}, E^o_{\text{in}}), \tag{20}$$
$$x_{10} = (b_r, b_{o,\text{echo}}, s, t, E^d_{\text{out}}, E^o_{\text{out}}), \tag{21}$$
$$x_{11} = (\text{Decode}_A(b_r), s, t), \tag{22}$$
$$x_{12} = (D_{\text{out}}, s, t, \text{Resolve}_A(D_{\text{out}}, s)). \tag{23}$$

Here the applicable data and operation crossing produces each displayed Wire state. The operation egress additionally requires $b_{o,\text{echo}} = b_o$ byte for byte. Manifestation accepts x_{12} only when

$$\text{Manifest}_{A,N}(D_{\text{out}}, s, t) = r \quad \text{and} \quad \text{ReadFinal}_N(t) = C_s. \tag{24}$$

Then $x_{13} = r$ and $x_{14} = \text{RouteResult}_{A,N}(r)$.

Each phase declaration fixes its identifier, index, exact X_i and X_{i+1} types, operator binding, and evaluator binding before execution. All displayed operations propagate typed rejection. Consumed values are released before the next transition unless explicitly present in the next product; no complete boundary-value vector is retained.

4 Independent bounded tensor section

For $Q = (q, b)$ and conforming profile R_A, the bounded section is

$$\mathrm{WindowSection}_A(D, Q; R_A) = \mathrm{Realize}_A(D, Q; R_A) = (W_q, K_q) \quad \text{or } \mathsf{REJECT}.$$

It binds the same epoch and canonical tensor identity as the relevant main route boundary and requires $0 \le |\beta_{A,\Lambda}(D|q)| \le b \le B_{\mathrm{window}}$. It produces exactly the window W_q and continuation set K_q. It does not change D, does not return $\mathrm{Resolve}_A(D, s)$, and does not feed $\mathrm{Manifest}_{A,N}$. Its proof may be referenced by identity while its values remain outside the main route.

Proposition 2 (Codomain separation)**.** *Substitution of W_q for C_s is ill typed. Binding the independent window section to D preserves both companion contracts without adding a conversion operator or alternate decoder.*

Proof. W_q is a layout-bound selected tensor-closure image, whereas $C_s \in U^*$ is the result of Ionizer selected mapback. Their defining equations and roles are distinct. The separate section uses each at its declared codomain. □

5 Optional proof section

A proof section exists only when a proof profile P is fixed in the route declaration. Its exact signatures are

$$\mathrm{Prove}_{A,P} : (\mathrm{Statement}_{A,P}, \mathrm{ProverInput}_{A,P}) \to \mathrm{ProofBytes}_{A,P} \sqcup \{\mathsf{REJECT}\}, \tag{25}$$

$$\mathrm{Verify}_{A,P} : (\mathrm{Statement}_{A,P}, \mathrm{ProofBytes}_{A,P}) \to \mathrm{VerifiedStatement}_{A,P} \sqcup \{\mathsf{REJECT}\}. \tag{26}$$

Without P, neither phase is present, neither a profile nor a prover value is part of X_0, and Close consumes X_{13} directly. With P, the exact suffix is

$$\begin{aligned} X_{14,P} &= \mathrm{ManifestReceipt}_{A,N} \times \mathrm{Statement}_{A,P} \times \mathrm{ProofBytes}_{A,P}, \\ X_{15,P} &= \mathrm{ManifestReceipt}_{A,N} \times \mathrm{VerifiedStatement}_{A,P}, \\ X_{16,P} &= \mathrm{RouteResult}_{A,N}\,. \end{aligned}$$

Prove derives the canonical statement from the route identity, manifestation link digest, and fixed claim, then consumes the prover input derived from the fixed profile; it is not a free caller value. Verify produces the verified statement, and Close consumes it with the receipt. The profile, derived prover input, and both proof phases are route-identity bound. Any partial, reordered, duplicated, or profile-free proof suffix rejects. The route binds the selected proof relation; it does not extend the soundness claim of the proof system or its setup [2].

6 Canonical identities and exact equality

For each phase type X, fix one canonical encoding. Define

$$\mathrm{type_id}_A(X) = H(\texttt{AION-TYPE-V2} \| A \| \,\mathrm{Canonical}(X)), \tag{27}$$

$$\mathrm{value_id}_A(X, x) = H(\texttt{AION-VALUE-V2} \| A \| \,\mathrm{type_id}_A(X) \| \,\mathrm{Canonical}_X(x)). \tag{28}$$

Exact phase and adjacency equality is byte equality of canonical typed values, or identity of the same immutable typed boundary object. Digest equality is not an equality proof. SHA-256 commitments provide mutation evidence only under their collision and second-preimage assumptions [6].

The derived route identifier binds the epoch, entrypoint, route vector, all phase declarations, the exact canonical operation operand, the declared data and operation Wire route slots, selector, target namespace, and optional proof profile. The concrete route and digest states produced at both crossings are present in the typed values and therefore in the applicable link commitments. The caller cannot replace any of those values with a phase label or outer opcode. Its exact domain prefix is `AION-ROUTE-V2`; route, type, value, link, chain-start, and link-digest bodies use only their stated V2 domains. A V1, V3, unversioned, or unhashed-type body is distinct and rejects.

7 Link construction

For admitted $\mathrm{Eval}_A[i](x_i) = \mathsf{OK}(x_{i+1})$, define the canonical link body

$$\begin{aligned} L_i = \mathrm{CanonicalRecord}(&\texttt{AION-TYPED-LINK-V2}, A, e_A, \rho_A, \mathrm{phase}_i, i, n,\\ &\mathrm{type_id}_A(X_i), \mathrm{type_id}_A(X_{i+1}), o_i, \eta_i, h_{i-1},\\ &\mathrm{value_id}_A(X_i, x_i), \mathrm{value_id}_A(X_{i+1}, x_{i+1}), \mathsf{OK}), \end{aligned}$$

with

$$h_{-1} = H(\texttt{AION-CHAIN-START-V2} \| \rho_A \| A \| e_A), \qquad h_i = H(\texttt{AION-TYPED-LINK-DIGEST-V2} \| L_i).$$

There is no extension field. L_i contains declarations and commitments, not a phase value, source handle, dictionary, pocket, source-record identifier, or restoration recipe.

8 Streaming verification

Link verification receives only the current exact input, current exact output, one link, and the validated predecessor. It strictly parses the record, checks all declarations, compares canonical bytes, freshly evaluates the declared relation, recomputes the value commitments, and recomputes the link digest. The evaluator binding is derived from the actual declared callable and checked with its fixed epoch, entrypoint, route identity, route input, and canonical phase position; callable or evaluator-environment substitution under an unchanged label rejects. Link and chain verification return verified evidence or typed rejection rather than Boolean truth.

Timeout, unavailability, callable-identity mismatch, exception, malformed input, and semantic failure are distinct typed rejection results. None creates a successful link or escapes the aggregate result algebra.

Whole-chain verification accepts $(C, D_{\mathrm{op}}, s, t)$, the ordered links, the terminal result, and expected terminal digest. It does *not* accept a boundary-value vector. It keeps at most x_i, x_{i+1}, h_{i-1}, and explicitly bounded operator state; after link i verifies, consumed x_i is released. Thus link evidence cannot become a retained restoration ledger.

Lemma 3 (Streaming continuity)**.** *If link i verifies and phase $i+1$ begins, the canonical typed input of phase $i+1$ equals the exact output of phase i.*

Proof. The verifier supplies the freshly evaluated x_{i+1} directly to the next phase and compares canonical bytes at the transition. The hash is recomputed for evidence but is not used as the equality premise. □

9 Presently verified section re-entry

A digest alone is not an anchor. Re-entry at position j requires a *VerifiedPrefixAnchor* containing the exact route, epoch, entrypoint, index $j-1$, boundary type X_j, boundary value x_j, predecessor digest, and a present prefix-verification receipt. An independently supplied certificate is admissible only after its exact route, epoch, boundary, and predecessor relation are currently verified. Orphan, stale, copied, cross-route, or cross-epoch anchors reject.

Proposition 4 (Section agreement)**.** *Verification from a presently verified prefix anchor returns the same terminal link digest as the corresponding whole-chain segment.*

Proof. Both executions start with the same exact boundary value and predecessor, then apply the same canonical link equation and fixed evaluator at every position. Induction over the section length gives the same digest sequence. □

10 Datatype and aperture boundary

The patent fidelity ladder

$$\mathsf{sint16} \to \mathsf{sint8} \to \mathsf{sint4} \to \mathsf{sint2} \to \mathsf{sint1}$$

is distinct from the raw observation-aperture ladder

$$1, 2, 4, 8, 16 \text{ bytes.}$$

No route phase has a universal signed-int16 premise. Clipping, saturation, wrapping, truncation, lossy quantization, collision acceptance, and silent cross-type coercion are excluded from the exact boundary.

11 Patent foundation locators

The signed-as-filed patent (SHA-256 `e2975c587ae6299f86a84e4630c51a4de19b72abfaf4ec45c853f68775418240`) provides technical provenance for the envelope and deterministic-transition foundation. Table 2 records the page, specification paragraph, and claim locators used here; the present definitions remain the phase-chain authority.

Table 2: Patent foundation locators.

Phase-chain leaf	Exact locator
Fixed route and transition identity	PDF pp. 8–9 / spec pp. 4–5, paras. [0023]–[0025]; PDF pp. 28–30 / spec pp. 24–26, paras. [0100]–[0109]; claims 12, 14, 19.
Canonical representation and fidelity	PDF pp. 10–11 / spec pp. 6–7, paras. [0029]–[0030]; PDF pp. 17–21 / spec pp. 13–17, paras. [0058], [0060]–[0070]; claims 2, 9, 10, 17, 18.
Validation and normal form	PDF pp. 18–19 / spec pp. 14–15, paras. [0060]–[0064]; claims 1, 2, 17.
Selected mapback	PDF pp. 22–24 / spec pp. 18–20, paras. [0076]–[0080]; PDF pp. 33–34 / spec pp. 29–30, paras. [0120]–[0124]; claims 8, 20.
Two-operand calculation	PDF pp. 27–28 / spec pp. 23–24, paras. [0094]–[0099]; claims 8, 11, 16, 20.
No-smuggling and fail-closed proof binding	PDF pp. 28–30 / spec pp. 24–26, paras. [0100]–[0109]; PDF pp. 21–26 / spec pp. 17–22, paras. [0071]–[0089]; claims 5–7, 12–15, 19, 20.

12 Main theorem

Theorem 5 (Exact typed phase-chain closure)**.** *Fix a valid epoch A, exact companion contracts, canonical operation tensor D_{op}, selector s, target t, fixed route, and well-typed deterministic phase declarations. Then:*

(1) every admitted transition has one canonical link body and digest;
(2) local verification accepts only its exact declared relation;
(3) whole-chain verification accepts only the fixed ordered route with exact canonical-byte continuity;
(4) deletion, insertion, reordering, duplication, cross-route or cross-epoch use, entrypoint or phase substitution, type or value mutation, operation operand, selector or target mutation, operator or evaluator mutation, terminal mutation, and trailing data reject;
(5) a failed phase produces no admitted aggregate chain;
(6) an accepted route returns exactly the result of Equations (1)–(10) and (24), with receipt B^*, epoch, selector, selected length, digest, and final bytes all equal to the dense Resolve result;
(7) the independent window section returns exactly its declared window and continuation-set pair and is never substituted for Resolve or manifestation;
(8) evidence alone cannot restore any phase value;
(9) a presently verified section agrees with whole-chain verification;
(10) exact equality is independent of digest equality;
(11) stored records, old receipts, caches, and prior success do not provide present acceptance; and
(12) no retained boundary vector or alternate mapback is admitted.

Proof. The table and phase equations prove exact composability. Wire data identity and Ionizer encode/decode identity preserve the data operand, operation operand, and result tensor. The two Wire route/digest states are separately typed and linked; exact operation egress proves that the second canonical operand completed its own unchanged crossing rather than bypassing transport. Resolve and Manifest use the same (D_{out}, s). The manifestation receipt must bind B^*, epoch A, selector s, target t, exact selected length and digest, and $\mathrm{ReadFinal}_N(t) = C_s$; exact final bytes are compared directly.

Canonical link determinism follows from fixed declarations, canonical field order, minimal scalar encodings, and deterministic digest input. Streaming verification establishes adjacency by byte equality and fresh evaluation. Induction over route position proves ordered continuity; every listed mutation changes a checked field, type, value, relation, position, predecessor, length, or terminal comparison. The verifier's two-value lifetime prevents a boundary ledger. Section agreement follows from the verified-prefix proposition. Codomain separation proves the independent Origamold clause. □

13 Failure matrix

14 Position relative to established constructions

Hash chains and Merkle-style commitments bind ordered evidence [5]; session types describe communication structure and compatibility [3]; verified codecs establish precise parser and encoder relations [1]; and state-machine replication studies ordered transitions under different fault models [4]. The present theorem is narrower: it fixes one heterogeneous tensor route, makes every companion signature explicit, separates exact equality from digest evidence, and excludes any retained inverse ledger.

Case	Rejecting boundary
Missing, extra, reordered, or duplicated link	Exact chain length and position
Wrong route, entrypoint, or epoch	Derived declaration binding
Wrong input or output type	Exact adjacent type equality
Changed operation tensor, selector, or target	Route identity and fresh relation
Canonical value mutation	Direct canonical-byte comparison
Hash collision offered as equality	Equality/hash separation
Stored OK with failed relation	Fresh evaluation
Substituted callable under unchanged evaluator label	Callable identity
Timeout, unavailable evaluator, exception, or semantic failure	Typed rejection algebra
FAKE, BOGUS, or merely well-typed alternate route	Exact fixed route vector
Wrong proof profile or partial proof suffix	Route-bound proof-profile grammar
Wrong B^*, epoch, selector, target, final length, digest, or bytes	Energizer receipt and dense Resolve equality
Boundary-value vector supplied as authority	Streaming verifier grammar
Orphan section digest	Present prefix-anchor verification
Window substituted for selected bytes	Codomain separation
Universal signed-int16 interpretation	Typed fidelity/aperture boundary
Calculation or manifestation rejection	No admitted aggregate result

15 Scope

The theorem establishes exact typed route closure and mutation-detecting evidence for the declared epoch and route. It does not establish objective truth, permission, confidentiality, availability, calculation-engine safety, network delivery, proof-system soundness beyond its premises, physical erasure, or correctness outside the route. Internal cross-repository synchronization is not a premise of the mathematics.

16 Conclusion

An exact phase chain is a typed composition, not a label sequence. The construction fixes every boundary, carries both canonical calculation operands through complete Wire routes, recomputes every relation, and uses hashes only as evidence. The main route and bounded tensor section now preserve the exact codomains of Ionizer, Wire, Origamold, and Energizer without inventing a bridge or alternate mapback.

A Candidate formal specification

The complete candidate formal source is distributed beside this paper as `exact-typed-phase-chain-formal-specification-candidate.md`. It is not embedded because companion identities and owner implementation equivalence remain to be frozen. The release packet records those open gates and makes no publication-readiness claim.

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